F1 RACE INTELLIGENCE

DASHBOARD & ANALYTICS PROJECT

A concise project report on Formula 1 race data, interactive Tableau visualizations, and a deployed Flask web experience.

6,464	17	24	10
Race records	Seasons analysed	Circuits represented	Teams represented

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Tableau Public • Flask • Python • GitHub • Render

01 | Executive Overview

The F1 Race Intelligence Dashboard presents Formula 1 race data through a set of visual analytics designed to make driver, team, tyre, reliability, and season patterns easier to explore. The project combines a Tableau dashboard with a web application for sharing the work.

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Project objectives

• Compare driver and constructor performance using points and podium-related views.

• Explore tyre-compound distribution and how usage changes across seasons.

• Investigate driver reliability through recorded DNF reasons.

• Show the geographical spread of host countries and race completion trends.

• Provide filters that support focused exploration by season, team, tyre compound, and DNF reason.

Primary deliverables

Interactive Tableau Public dashboard and story, a Flask-based website, source code hosted on GitHub, and this concise project report. The supplied visuals are included as compact evidence of the actual Tableau views.

Project links

Website: https://f1-race-intelligence-mhgm.onrender.com/

GitHub repository: https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

Tableau dashboard:
https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD

Tableau story: https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence

02 | Solution & Technical Design

The solution separates data exploration from the web delivery layer. Tableau provides the interactive analytics experience, while the Flask application offers a browser-accessible project entry point.

Race data Structured records and fields	→	**Tableau analysis** Calculated measures, filters, charts	→	**Insights** Driver, team, tyre, reliability
Source code Project files	→	**GitHub** Versioned repository	→	**Render** Deployed Flask website

Technology stack

Layer	Technology / role
Data visualisation	Tableau Public — dashboards, charts, filters, story presentation.
Web application	Python and Flask — web routes and page delivery.
Version control	GitHub — repository and source-code management.
Deployment	Render — hosted website endpoint.
Data warehouse	Snowflake usage was not confirmed in the supplied project evidence; it is not claimed as part of the implemented stack.

Data source URL and a formal data dictionary were not supplied in the project materials. Metric definitions should therefore be documented alongside the final submission if required by the evaluator.

03 | Dashboard at a Glance

The dashboard brings together summary indicators, driver and team comparisons, tyre usage, podium achievements, host-country geography, completion-rate analysis, season-wise championship performance, and driver reliability.

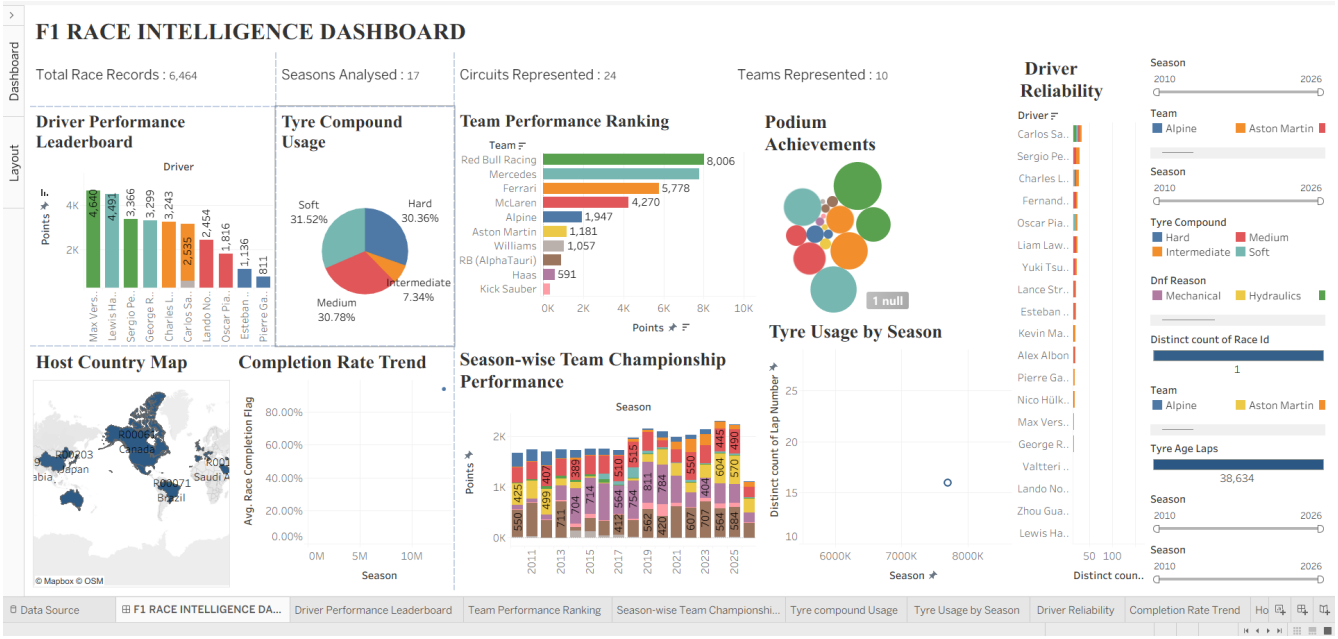


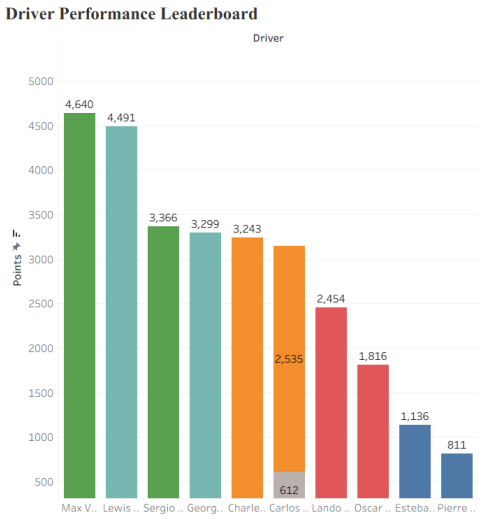
Figure 1. Supplied Tableau dashboard screenshot. The visual overview is shown at reduced size to preserve readability while keeping the report concise.

View	Analytical purpose
Driver leaderboard	Compare points totals among leading drivers.
Team ranking	Compare aggregate points across constructors.
Tyre usage	Show distribution of Soft, Medium, Hard, and Intermediate compounds.
Reliability and completion	Explore recorded DNF reasons and race completion patterns.
Season-wise views	Compare team championship points and tyre use over time.

04 | Performance & Championship Views

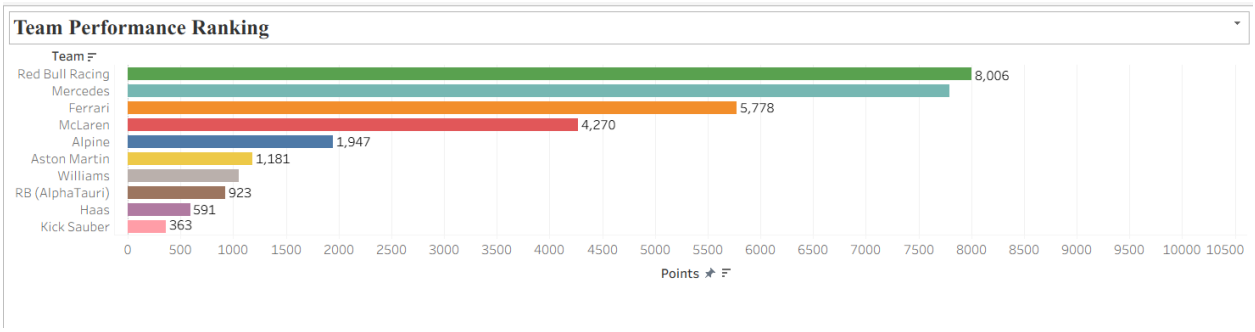
These views focus on competitive performance: individual driver points, team-level points, and the changing distribution of points across seasons.

Driver Performance Leaderboard



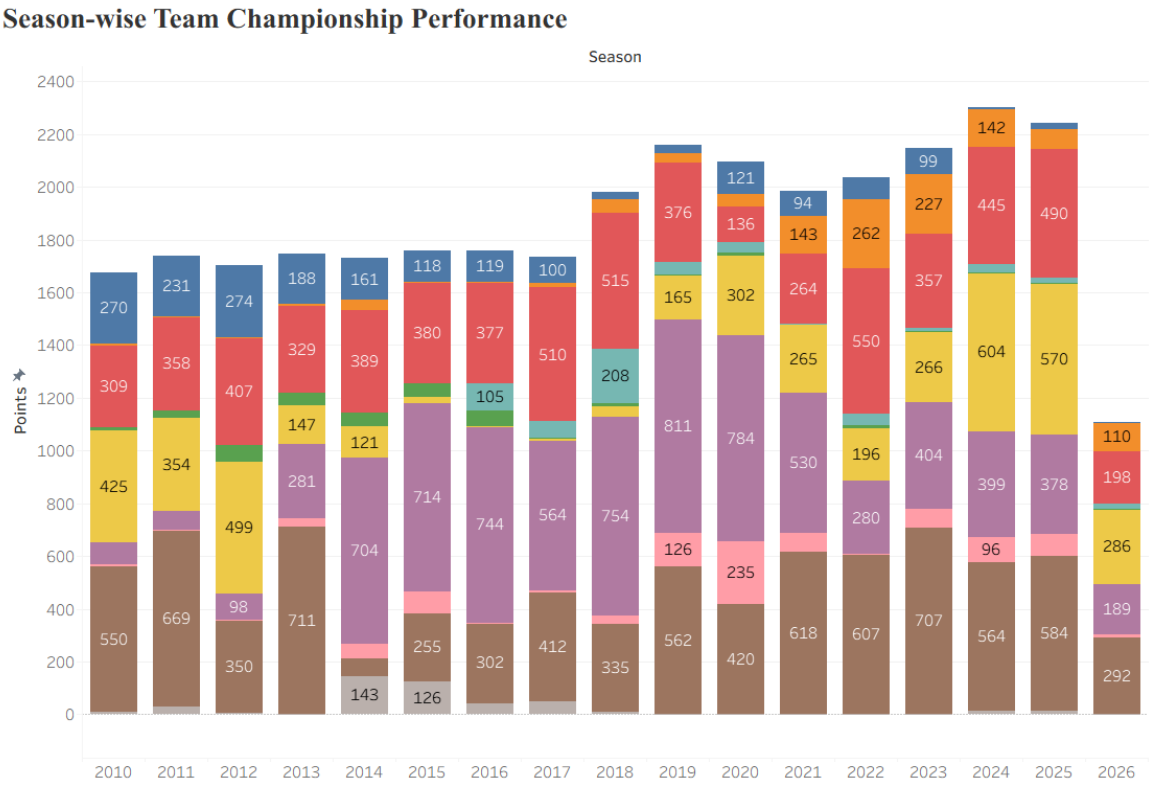
The leaderboard ranks the drivers shown by their displayed points totals. Driver names are shortened in the chart due to available width.

Team Performance Ranking



Red Bull Racing and Mercedes lead the displayed team ranking, followed by Ferrari and McLaren. These are the values shown in the supplied screenshot.

Season-wise Team Championship Performance



The stacked chart compares season totals by team. Its multi-colour segments allow the viewer to inspect how team point contributions change over the displayed seasons.

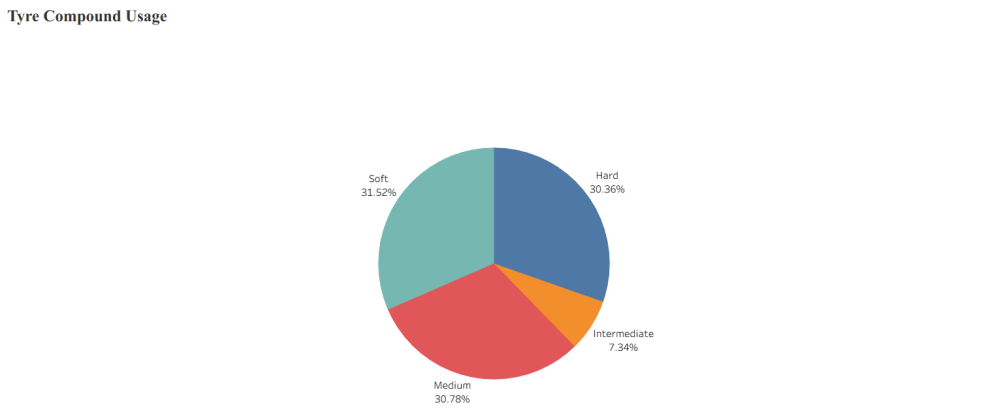
Interpretation note

Before presenting final conclusions, verify the aggregation rules, season coverage, and whether every team is represented consistently in each season. A chart should be described as a points comparison only where its underlying measure is confirmed to be points.

05 | Tyres, Podiums & Reliability

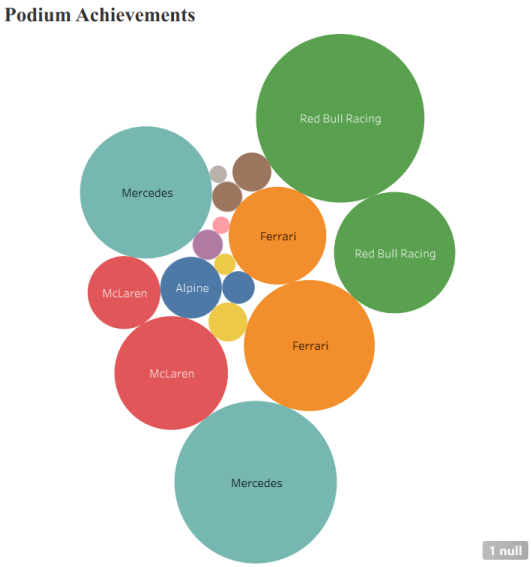
This group of visualizations supports analysis of tyre strategy, team podium-related records, and the reasons races end without a classified finish.

Tyre Compound Usage



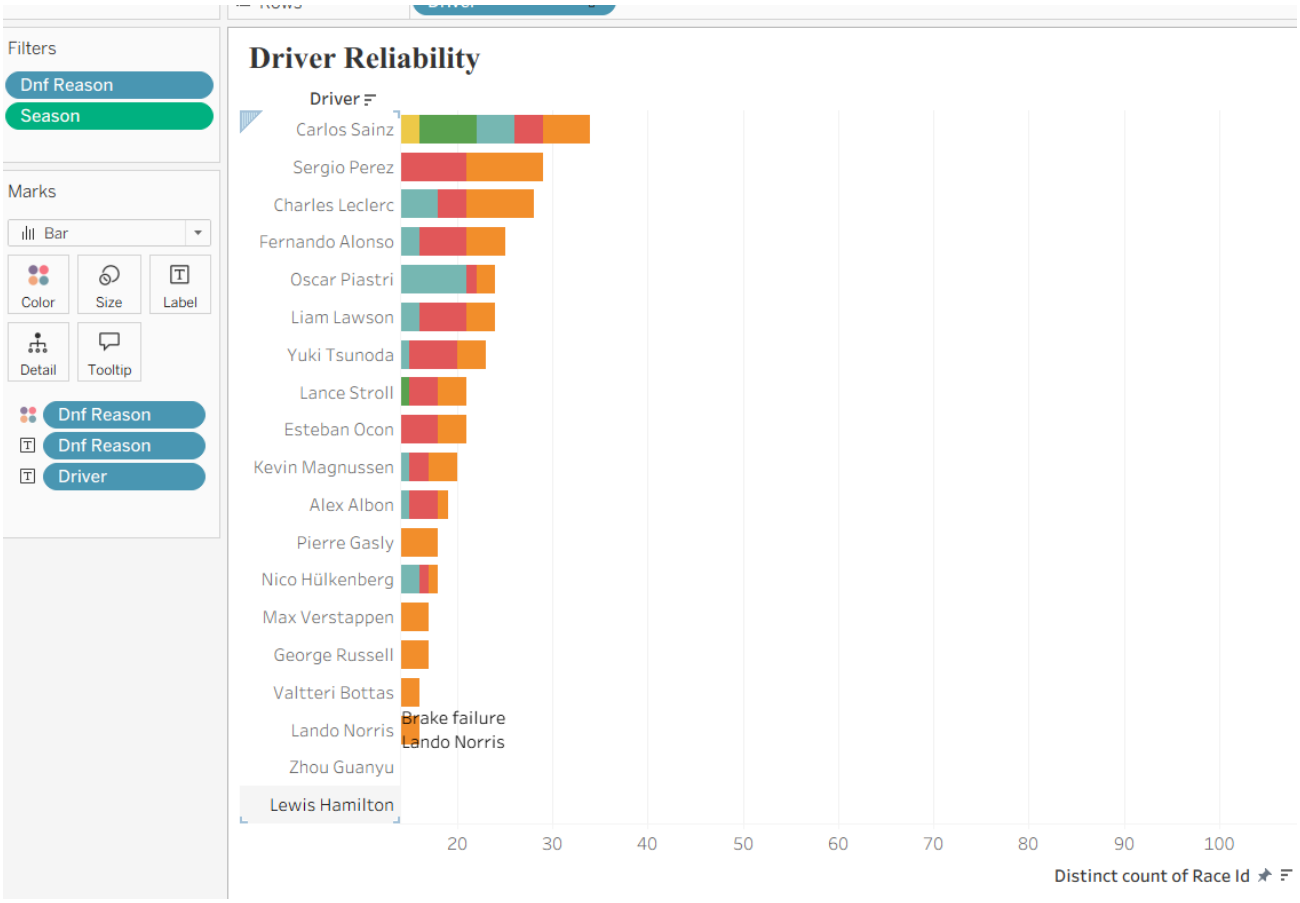
The supplied chart shows Soft at 31.52%, Medium at 30.78%, Hard at 30.36%, and Intermediate at 7.34%. Soft, Medium, and Hard account for similar shares in this view.

Podium Achievements



The bubble view compares the displayed team categories by bubble size. The screenshot includes a “1 null” indicator, so null or missing category values should be checked before final interpretation.

Driver Reliability



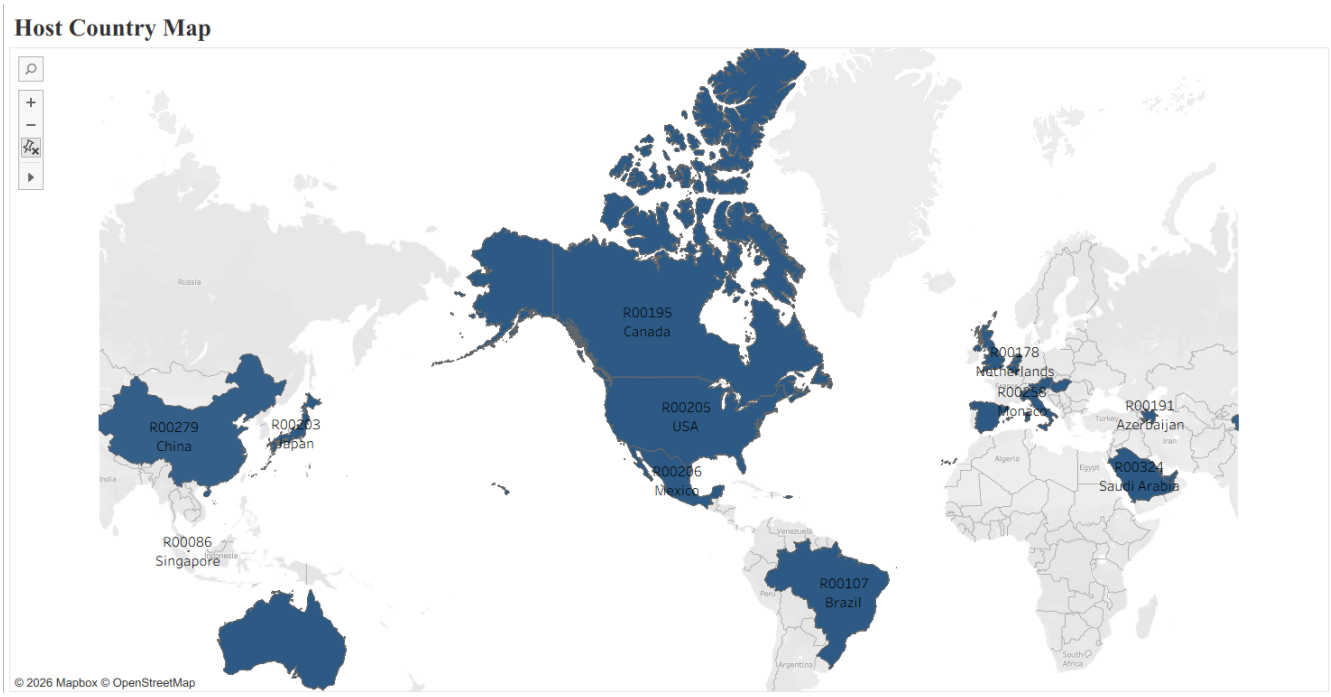
The stacked bars show the distribution of recorded DNF reasons for drivers in the view. The filters shown allow analysis by season and DNF reason.

Quality checks to complete

• Confirm that DNF reasons are consistently classified and that blank values are handled deliberately.
• Confirm whether the measure counts distinct Race IDs and whether the current aggregation matches the intended question.
• Use neutral wording: the view describes recorded retirements, not necessarily a driver's overall reliability without exposure or race-start context.

06 | Geography, Testing & Delivery

Host Country Map



The map locates host countries represented in the race data. It should not be interpreted as a map of driver nationalities.

Testing and validation checklist

Check	What to verify
Data quality	Check nulls, duplicates, valid seasons, driver/team labels, and category consistency.
Metric correctness	Validate point totals, podium measure, DNF counts, tyre percentages, and completion-rate formula.
Interaction	Test each filter independently and in combination; confirm charts update consistently.
Presentation	Check truncated labels, axis titles, null indicators, and any unexplained isolated marks.
Deployment	Open the hosted website and confirm all routes, links, and embedded Tableau views work.

Demonstration plan

1. Introduce the objective and dataset scope.
2. Walk through the summary metrics and dashboard layout.
3. Demonstrate season, team, tyre, and DNF filters.
4. Explain two or three observations visible in the charts.
5. Show the deployed website and GitHub repository, then acknowledge limitations and planned improvements.

07 | Future Plan & Conclusion

Recommended improvements

• **Data transparency:** add a source citation, data dictionary, refresh date, and definitions for every metric.

• **Visual polish:** expand truncated driver labels, remove unexplained nulls, and correct chart axes or aggregation where required.

• **Analytical depth:** add season-specific comparisons, race-start-adjusted reliability measures, and clear podium definitions.

• **Performance:** reduce unnecessary data volume, use efficient calculations, and test the website and dashboard on desktop and mobile layouts.

• **Accessibility:** retain meaningful titles, use readable contrast, and avoid relying on colour alone to communicate categories.

Scalability considerations

The project can be extended by adding more seasons, consistent team and driver mappings, documented data-refresh steps, and a repeatable validation checklist. Future changes should preserve the existing Tableau interactions while keeping the website and repository easy to maintain.

Conclusion

The F1 Race Intelligence Dashboard combines a shareable web presence with visual analysis of Formula 1 performance and race patterns. The supplied screenshots provide concrete evidence of the main views. Before final submission, the most important remaining step is to validate measure definitions and resolve visible quality flags such as null categories and truncated labels.

PROJECT LINKS

Website • f1-race-intelligence-mhgm.onrender.com
GitHub • F1-Race-Intelligence
Tableau dashboard • Open dashboard
Tableau story • Open story